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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Troubleshooting Network Services

Troubleshoot DNS and DHCP issues in a simulated network.

Task: Document solutions for common DNS and DHCP issues.

❖ **1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.**

➢ **Aim**

- To simulate a DNS resolution failure on a Windows computer by modifying the Hosts file, assigning an incorrect IP address to www.spotify.com, and testing the domain using the ping command.

➢ **Introduction**

- DNS (Domain Name System) is a network service that translates human-readable domain names into IP addresses.
- For example, when a user enters www.spotify.com in a web browser, DNS helps find the IP address associated with that domain.

- The Hosts file is a local file in Windows that maps domain names to IP addresses. Windows checks this file before using external DNS servers.
- By adding an incorrect IP address to the Hosts file, we can simulate a DNS resolution problem and observe how the computer responds.

➢ **Requirements**

1. Windows computer or laptop.
2. Administrator access.
3. Notepad application.
4. Command Prompt (CMD).
5. Internet connection (for normal DNS testing).

➢ **Procedure / Steps**

➢ **Step 1: Open Notepad as Administrator**

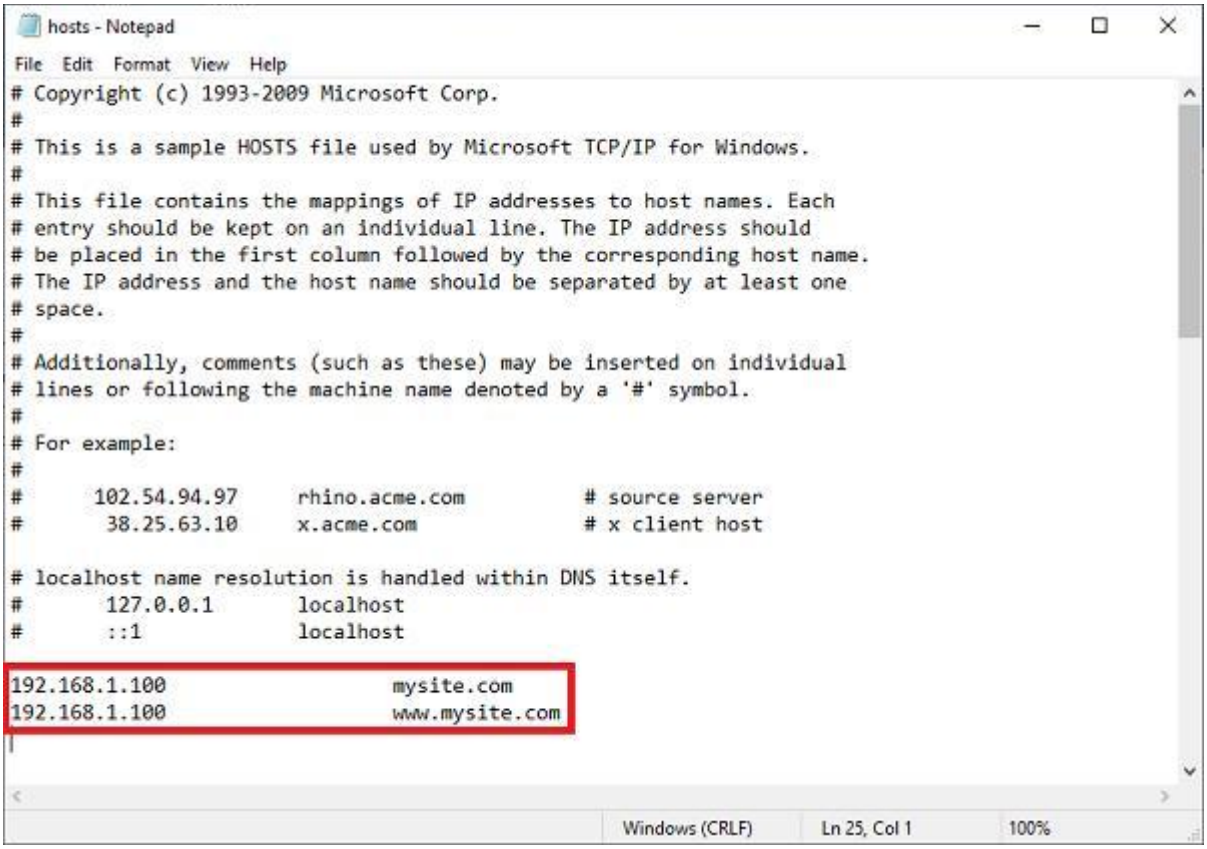
- Click on the Windows Start menu.
- Search for Notepad.
- Right-click on Notepad.
- Select Run as administrator.
- Click Yes if the User Account Control window appears.

➢ **Step 2: Open the Hosts File**

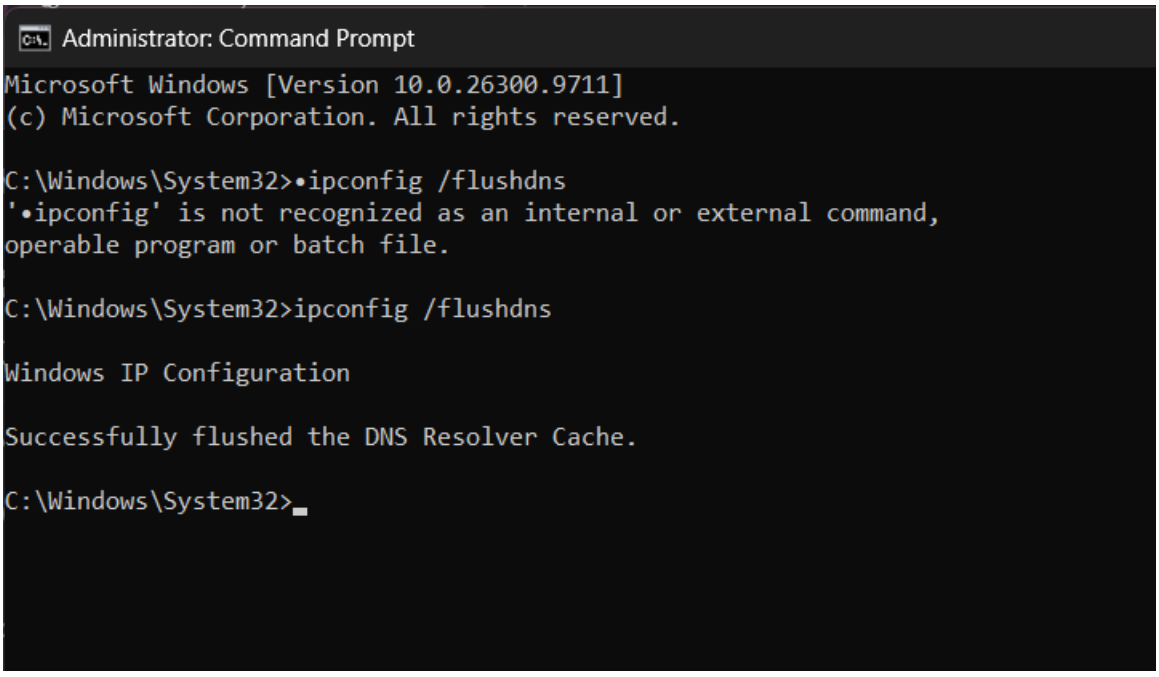
- In Notepad, click on File → Open.
- Navigate to the following location:
- C:\Windows\System32\drivers\etc
- Change the file type from Text Documents to All Files.
- Select the file named hosts.
- Click Open.

➢ **Step 3: Edit the Hosts File**

- Go to the bottom of the Hosts file.
- Add the following entry on a new line:
- 203.0.113.10 www.spotify.com
- Here, 203.0.113.10 is a reserved documentation IP address used for this simulation, not Spotify's actual server IP.
- This entry forces Windows to resolve www.spotify.com to the incorrect IP address.
- Save the file using Ctrl + S.



- **Step 4: Open Command Prompt**
  - Press Windows + R.
  - Type cmd.
  - Press Enter.
  - Command Prompt will open.
- **Step 5: Clear the DNS Cache**
  - Enter the following command:
  - ipconfig /flushdns
  - Press Enter.
  - This command clears the locally stored DNS cache.
  - It helps ensure that the new Hosts file entry is used during testing.



- **Step 6: Ping the Domain Name**
  - Enter the following command:
  - **ping www.spotify.com**
  - Press Enter.
  - Windows will use the incorrect IP address specified in the Hosts file.
  - The ping request will attempt to reach that IP address.
- **Step 7: Observe the Error Message**
- After executing the ping command, the output may look like this:

  Pinging www.spotify.com [203.0.113.10] with 32 bytes of data:

  Request timed out.
  Request timed out.
  Request timed out.
  Request timed out.

  Ping statistics for 203.0.113.10:
  Packets: Sent = 4, Received = 0,
  Lost = 4 (100% loss)
- Explanation:
  - The computer resolves www.spotify.com to the incorrect IP address 203.0.113.10.
  - The ping command attempts to send packets to that address.
  - Since the address is reserved for documentation and is not Spotify's actual server, the requests generally receive no reply.
  - Therefore, the output may show Request timed out.

```
C:\Users\Siddesha_G_V>ping 192.168.29.100

Pinging 192.168.29.100 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.29.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Siddesha_G_V>
```

- **Step 8: Restore the Original Hosts File**
  - Open the Hosts file again using Notepad as Administrator.
  - Find the entry:
  - **203.0.113.10 www.spotify.com**
  - Delete this line.
  - Save the file using Ctrl + S.
  - Open Command Prompt and execute:
  - ipconfig /flushdns
- The incorrect mapping will now be removed, and Windows can use normal DNS resolution again.

- **5. Observation Table**

Test	Observation
Before editing Hosts file	Domain uses normal DNS resolution.
After editing Hosts file	Domain points to incorrect IP address.
Ping command	ping www.spotify.com
Resolved IP	203.0.113.10
Error message	Request timed out
Packet loss	Generally 100%
After removing entry	Normal DNS resolution is restored.

- **Result**
- Successfully simulated a DNS-related network problem by modifying the Windows Hosts file and assigning an incorrect IP address to www.spotify.com. After executing the ping command, the domain resolved to the incorrect IP address, and the ping requests timed out.

❖ **2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.**

➢ **Aim**

- To configure a DHCP server in Cisco Packet Tracer, intentionally assign an incorrect default gateway, connect a client device, verify the assigned gateway, and troubleshoot the issue by correcting the DHCP configuration.

➢ **Introduction**

- DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network configuration details to devices connected to a network.

➢ These configuration details include:

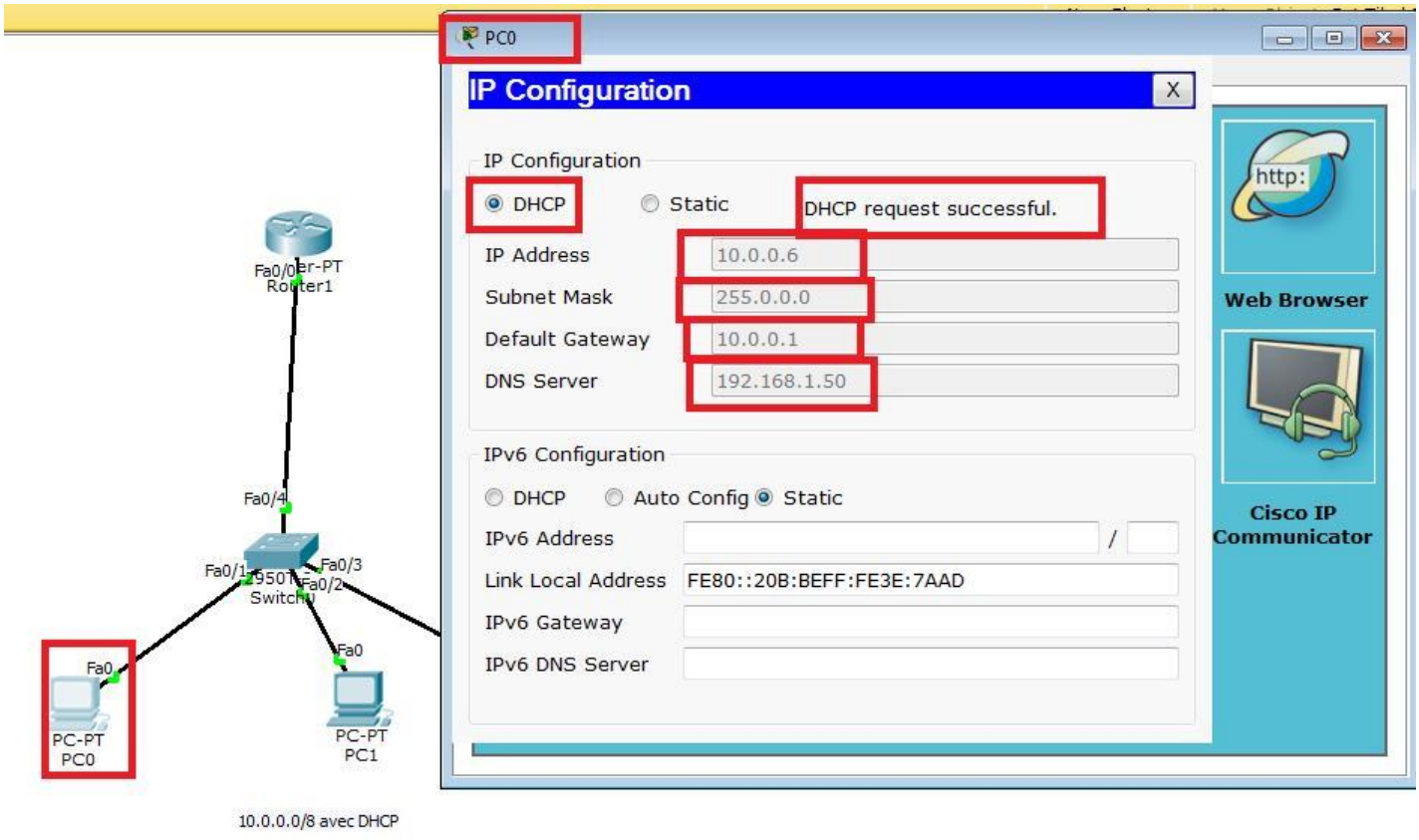
- IP Address
- Subnet Mask
- Default Gateway
- DNS Server Address

➢ A Default Gateway is the IP address of a router that allows devices to communicate with other networks, such as the internet.

➢ If a DHCP server provides an incorrect default gateway, the client may receive a valid IP address but fail to communicate with devices outside its local network.

- **Requirements**
  1. Cisco Packet Tracer software.
  2. One Router.
  3. One Server (DHCP Server).
  4. One Switch.
  5. One PC (Client Device).
  6. Ethernet cables.

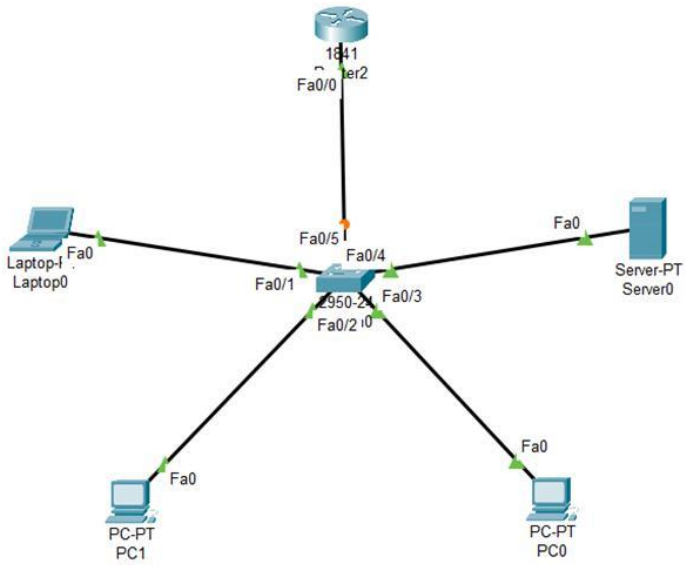
- **Network Topology**
- Create the following network topology in Cisco Packet Tracer:



- Network Configuration:

Device	IP Address
Router	192.168.1.1
DHCP Server	192.168.1.2
Client PC	Automatically assigned by DHCP
Subnet Mask	255.255.255.0
Correct Default Gateway	192.168.1.1
Incorrect Default Gateway	192.168.1.254

- **Procedure / Steps**
- **Step 1: Create the Network Topology**
- Open Cisco Packet Tracer.
- Add one Router, one Switch, one Server, and one PC.
- Connect the Router, Server, and PC to the Switch using Copper Straight-Through cables.
- Make sure all devices are properly connected.



- **Step 2: Configure the Router**
- Click on the Router.
- Open the CLI tab.
- Enter the following commands:

  enable
  configure terminal
  interface gigabitEthernet 0/0
  ip address 192.168.1.1 255.255.255.0
  no shutdown
  exit
  end
- The router is now configured with IP address 192.168.1.1.

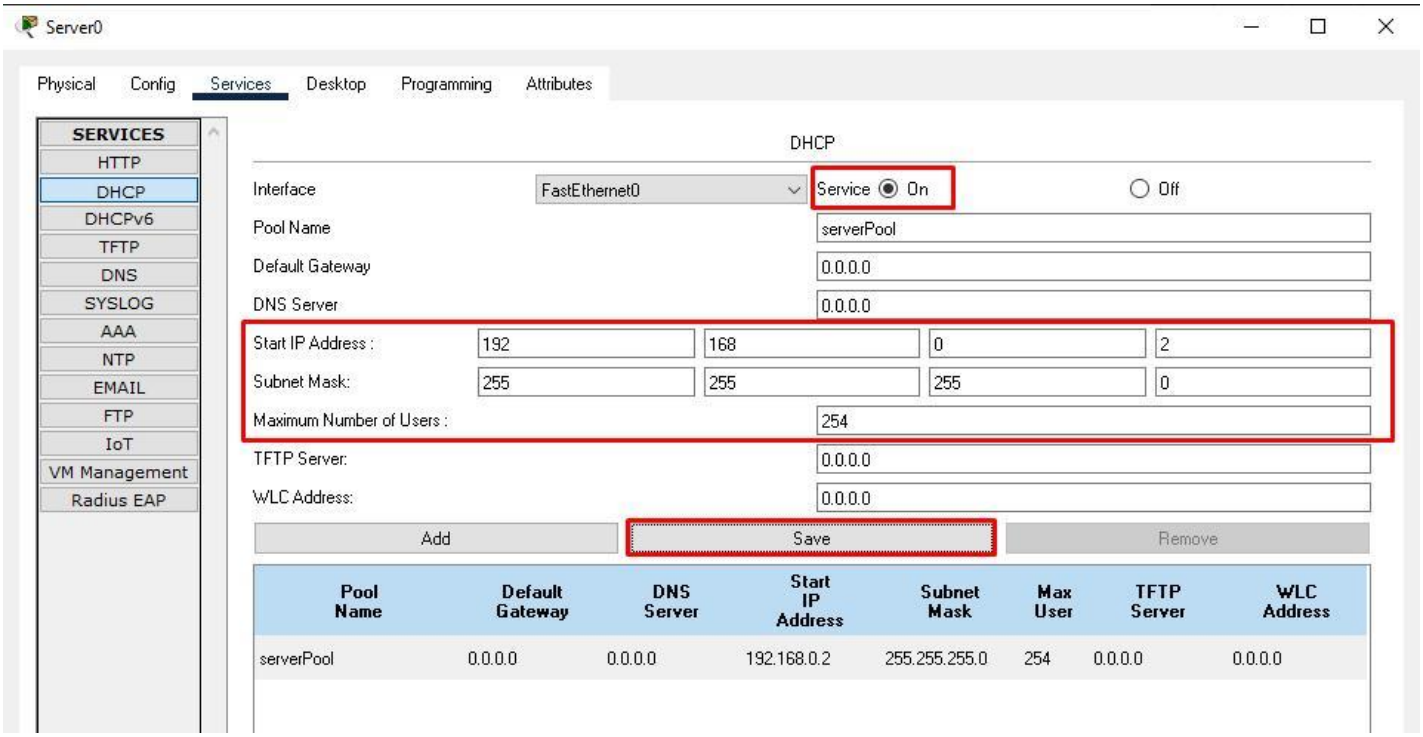
- **Step 3: Configure the DHCP Server**
  - Click on the Server.
  - Open the Desktop tab.
  - Select IP Configuration.
  - Set the following static configuration:

Setting	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

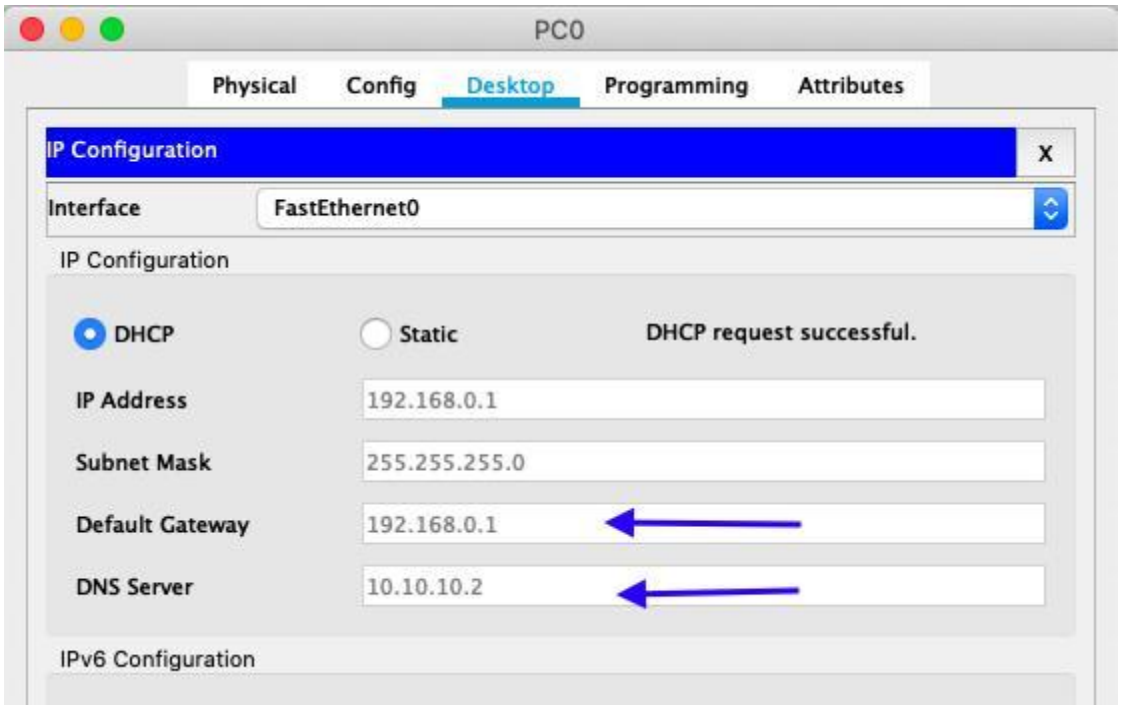
- Now open the Services tab.
- Select DHCP.
- Turn the DHCP service ON.
- Enter the following DHCP pool details:

DHCP Setting	Value
Pool Name	LAN_POOL
Default Gateway	192.168.1.254
DNS Server	8.8.8.8
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

- - Click Add or Save, depending on the Packet Tracer version.



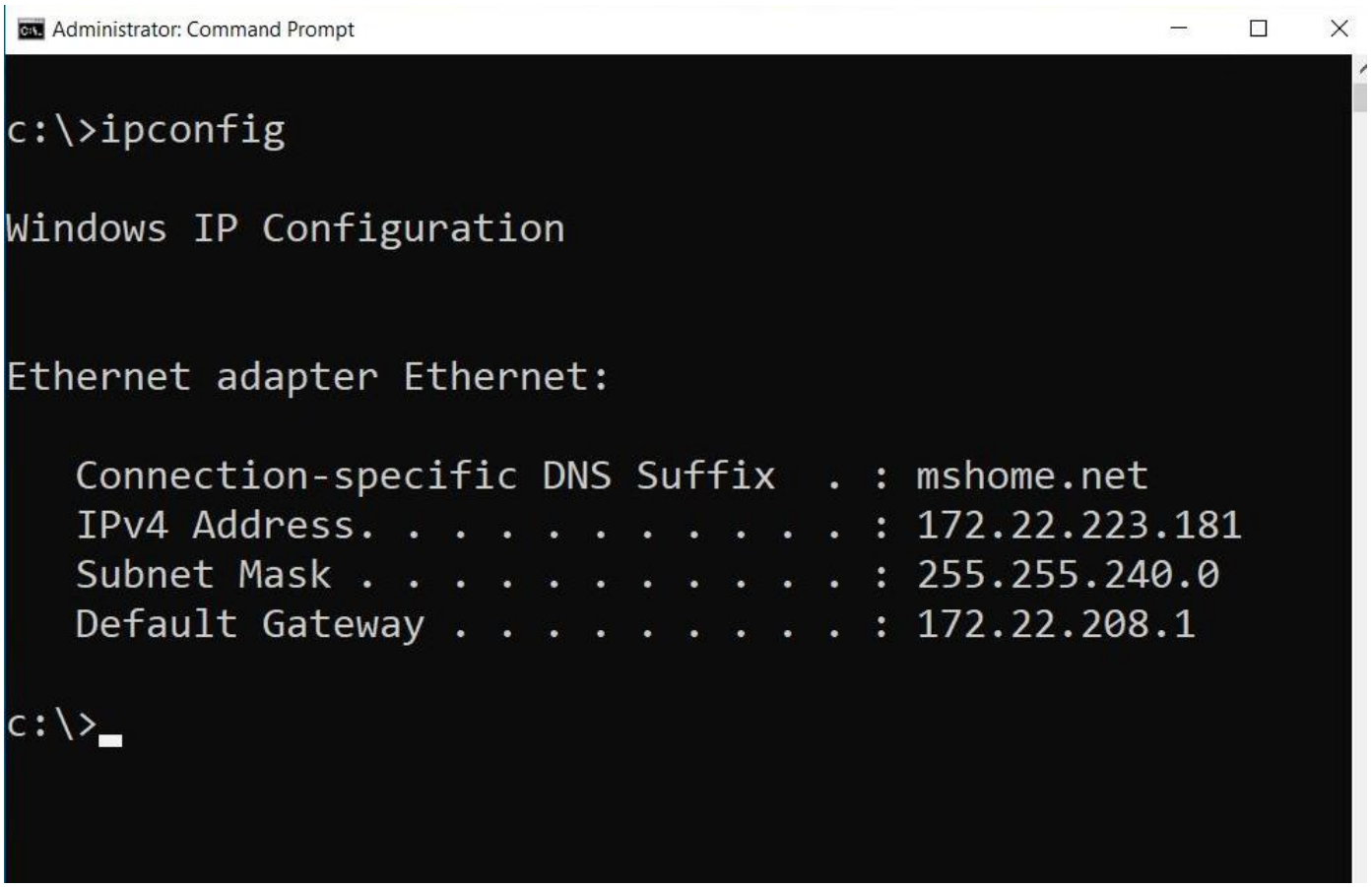
- **Step 4: Configure the Client PC**
- Click on the PC.
- Open the Desktop tab.
- Select IP Configuration.
- Click on DHCP.
- The PC will automatically request network configuration from the DHCP server.
- The client will receive an IP address, subnet mask, DNS server, and incorrect default gateway.



- **Step 5: Verify the Assigned Default Gateway**
- Open the PC's Desktop tab.
- Select Command Prompt.
- Enter the following command:
- **ipconfig /all**
- Press Enter.
- Check the assigned IP address, subnet mask, and default gateway.
- Expected Output (Example):

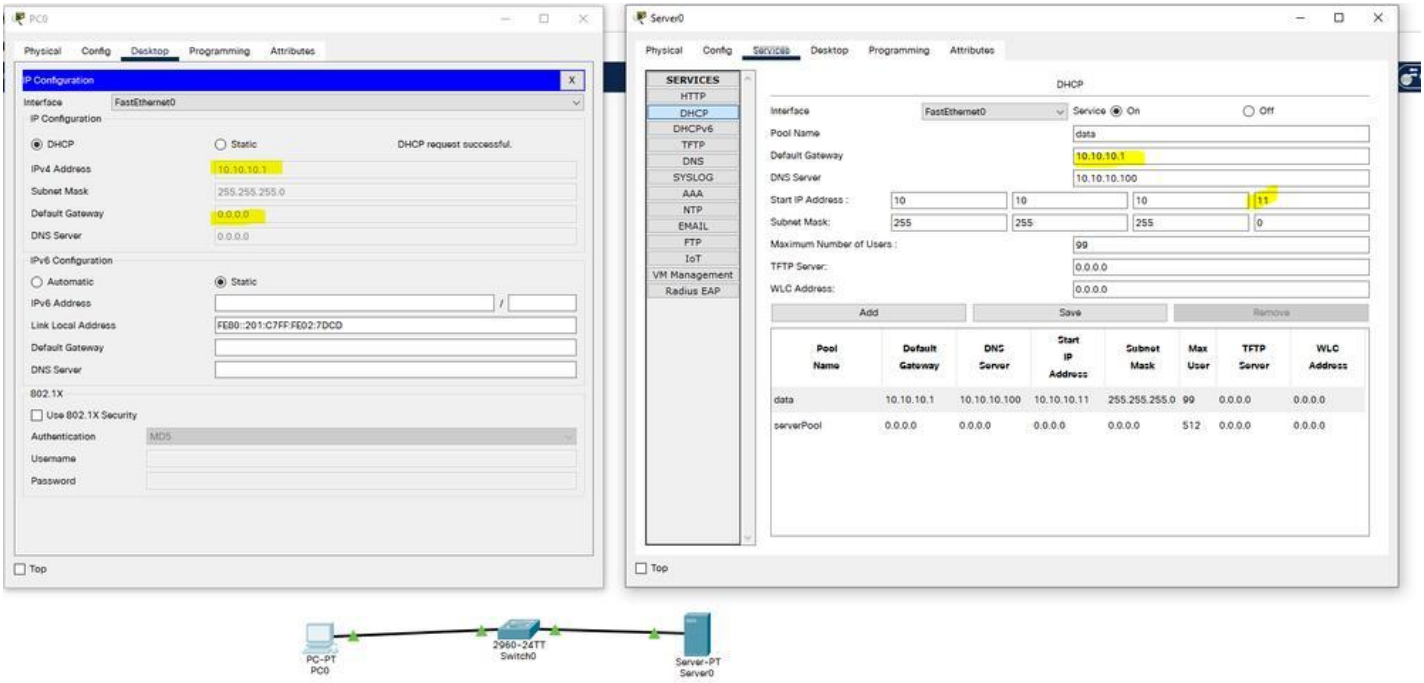
  IP Address: 192.168.1.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.1.254
  DNS Server: 8.8.8.8

- This confirms that the DHCP server has assigned the incorrect default gateway to the client.

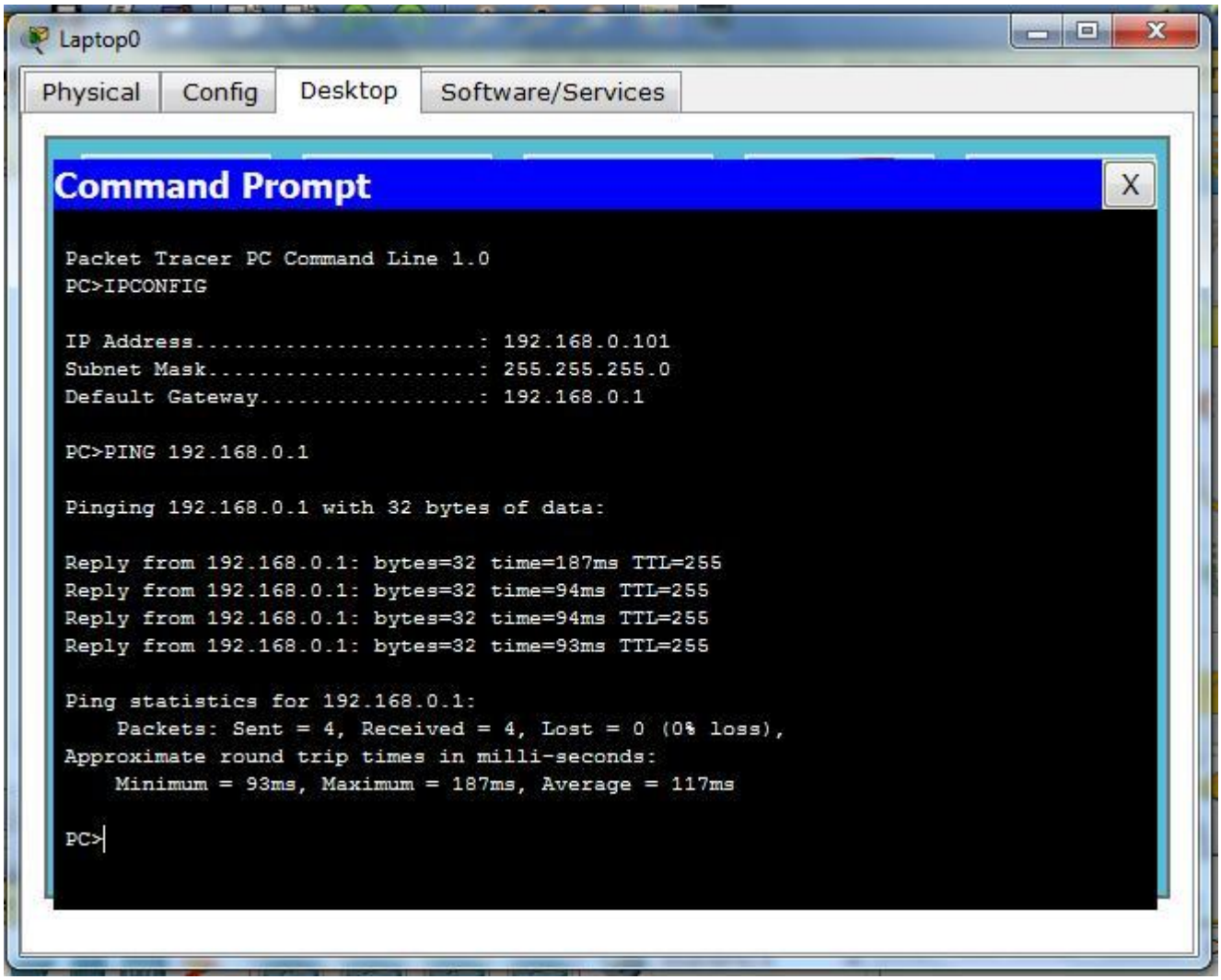


- **Step 6: Identify the Network Problem**
- Open the PC's Command Prompt.
- Ping the router using the following command:
  ping 192.168.1.1
- The ping should work because the router is on the same local network.
- - Now try to communicate with an external network using:
  ping 8.8.8.8

- The external ping may fail because the client has an incorrect default gateway and cannot properly forward traffic to other networks.
- Possible Error Message:
- Request timed out.
- This indicates a possible network connectivity problem. The incorrect default gateway is identified by comparing the DHCP-assigned gateway with the router's actual IP address.
- **Step 7: Fix the Incorrect Default Gateway**
  - Open the DHCP Server.
  - Go to Services → DHCP.
  - Select the configured DHCP pool.
  - Change the Default Gateway from:
- 192.168.1.254
- To the correct gateway:
- 192.168.1.1



- **Step 8: Renew the Client IP Configuration**
- Open the Client PC.
- Go to Desktop → IP Configuration.
- Click Static and then DHCP again, or use the following commands in Command Prompt:
- ipconfig /release
- ipconfig /renew
- - Check the assigned gateway again:
- ipconfig /all
- The client should now receive the correct default gateway 192.168.1.1.
- - Test the connection again using:
- ping 192.168.1.1
- The router should respond successfully.



- **Observation Table**

Parameter	Before Fixing	After Fixing
IP Address	192.168.1.10	192.168.1.10
Subnet Mask	255.255.255.0	255.255.255.0

Default Gateway	192.168.1.254	192.168.1.1
Gateway Configuration	Incorrect	Correct
Network Communication	External communication may fail	External communication can work if routing is configured correctly

- **Result**
- Successfully configured a DHCP server in Cisco Packet Tracer and intentionally assigned an incorrect default gateway. The client device received the incorrect gateway automatically. After identifying and correcting the DHCP configuration, the client received the correct default gateway.

❖ **3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.**

➢ **Aim**

- To diagnose and troubleshoot a DNS resolution problem in a simulated network where devices can access websites using IP addresses but cannot access them using domain names, with the help of the nslookup command.

➢ **Introduction**

- DNS (Domain Name System) is a network service that converts human-readable domain names into IP addresses.
- For example, when a user enters www.google.com in a web browser, DNS translates the domain name into its corresponding IP address.
- Sometimes, devices can access websites using their IP addresses but cannot access them using domain names. This usually indicates a DNS configuration or DNS resolution problem.
- The nslookup tool is used to check whether a DNS server can successfully resolve a domain name into an IP address.

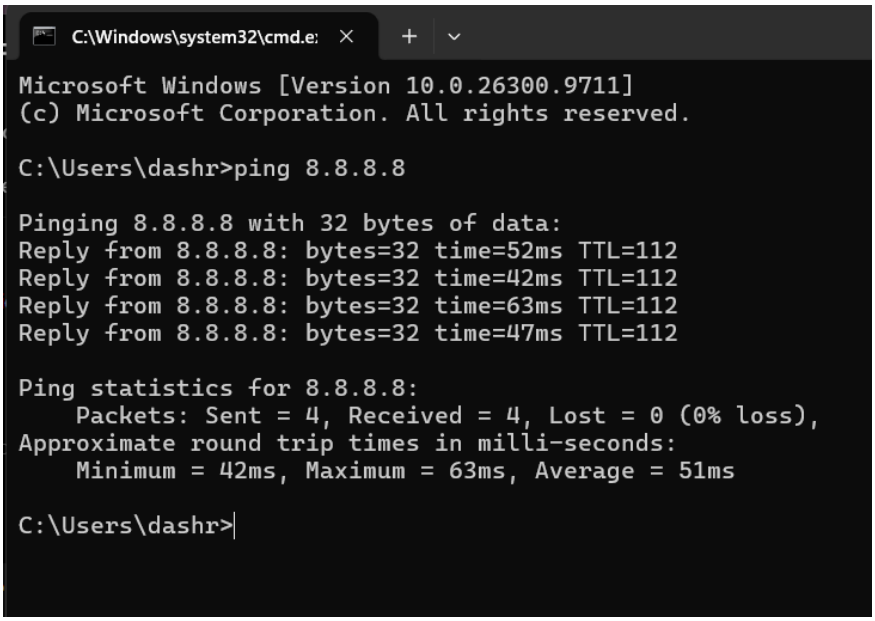
➢ **Requirements**

1. Computer or laptop.
2. Cisco Packet Tracer or GNS3 (for simulated network).
3. Client PC connected to the network.

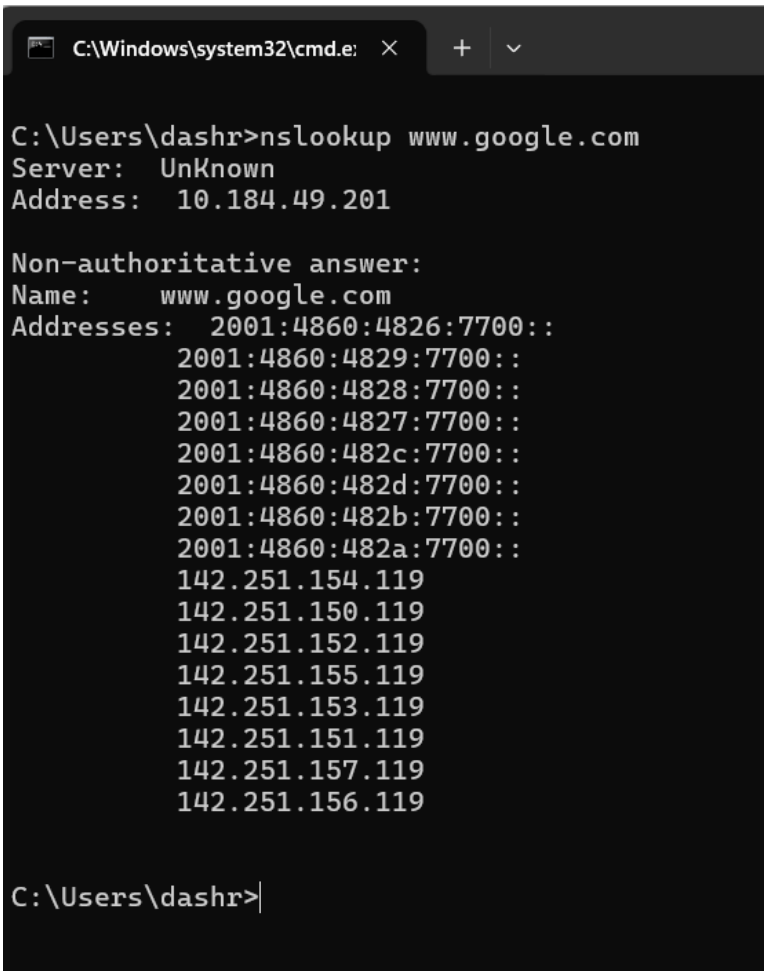
4. DNS server.
5. Command Prompt (CMD).
6. Internet connection (for testing public domains, if required).

- **Problem Scenario**
- Suppose a computer is connected to a network and can access a website using its IP address, but cannot open the same website using its domain name.
- For example:
  - Accessing a website using its IP address: Successful.
  - Accessing the website using its domain name: Failed.
- This indicates that the network connection may be working, but the DNS server is not resolving domain names correctly.

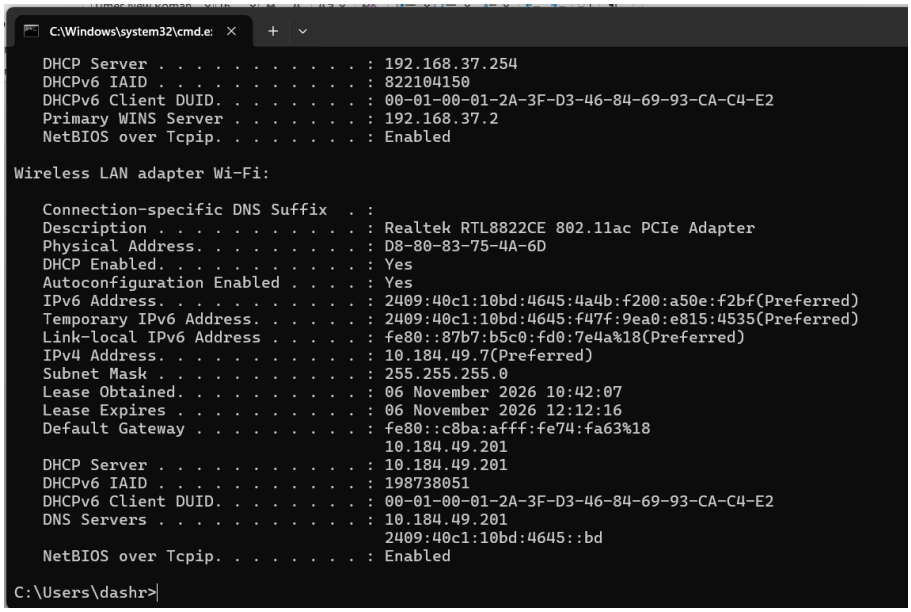
- **5. Procedure / Troubleshooting Steps**
- **Step 1: Check Network Connectivity**
- Open Command Prompt on the client computer.
- Enter the following command:
- ping 8.8.8.8
- Press Enter.
- Check whether the destination IP address responds.



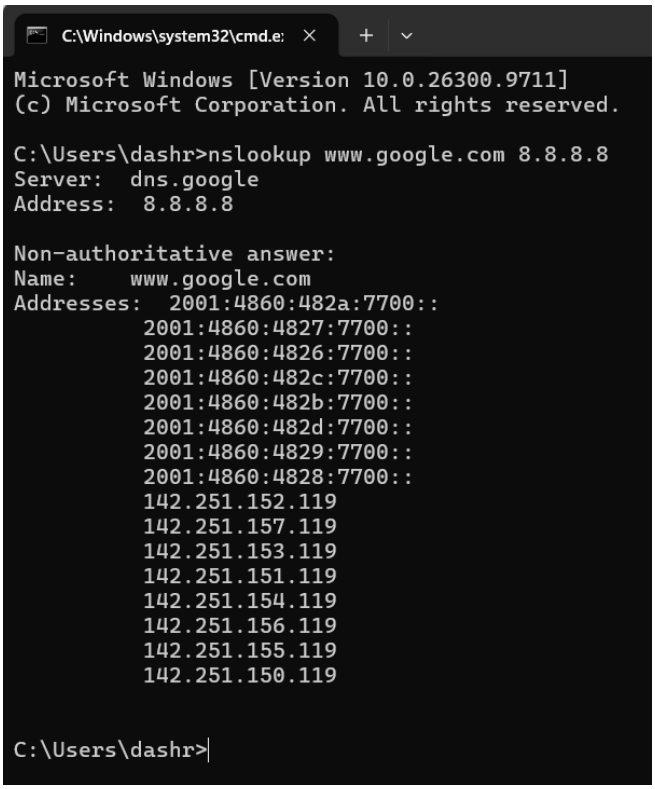
- **Step 2: Test Domain Name Resolution**
- - In Command Prompt, enter:
- nslookup www.google.com
- Press Enter.



- **Step 3: Check the DNS Server Configuration**
- - Execute the following command:
- ipconfig /all
- Check the DNS Servers field.
- Verify whether the client is using the correct DNS server IP address.
- If the DNS server address is incorrect or unavailable, domain name resolution may fail.



- **Step 4: Test the DNS Server Directly**
- Use the following command to query a specific DNS server:
  nslookup www.google.com 8.8.8.8
- Press Enter.
- This command sends the DNS query directly to Google's public DNS server.
- If this query succeeds but the default nslookup fails, the problem is likely with the configured DNS server.
- Example Successful Output:



- **Step 5: Identify the DNS Problem**
- After performing the above tests, compare the results:

Test	Result	Observation
Ping 8.8.8.8	Successful	IP connectivity is working.
nslookup www.google.com	Failed	DNS resolution problem exists.

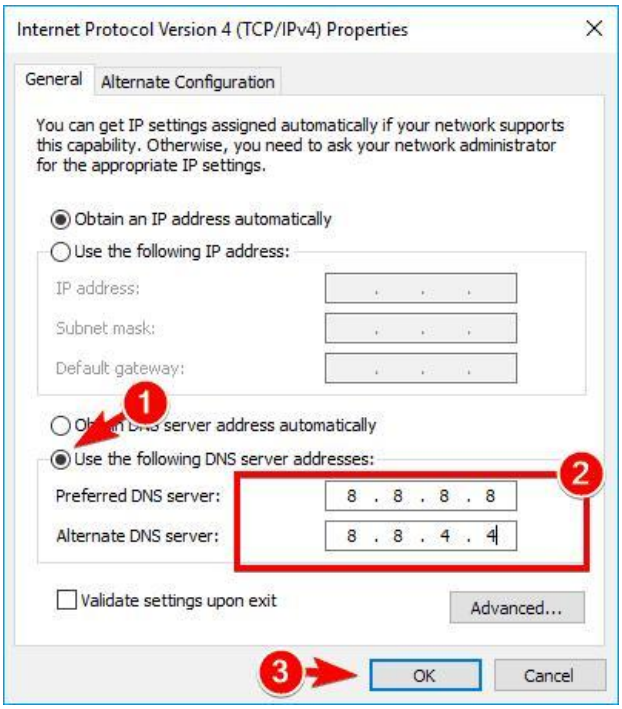
Check DNS configuration	Incorrect DNS IP	Client may be using an invalid DNS server.
nslookup using 8.8.8.8	Successful	Public DNS server is responding.

- **Step 6: Fix the DNS Problem**
- To resolve the DNS issue, follow these steps:

1. Open the network adapter settings.
2. Go to Internet Protocol Version 4 (TCP/IPv4) properties.
3. Select Use the following DNS server addresses.
4. Enter the following DNS server details:

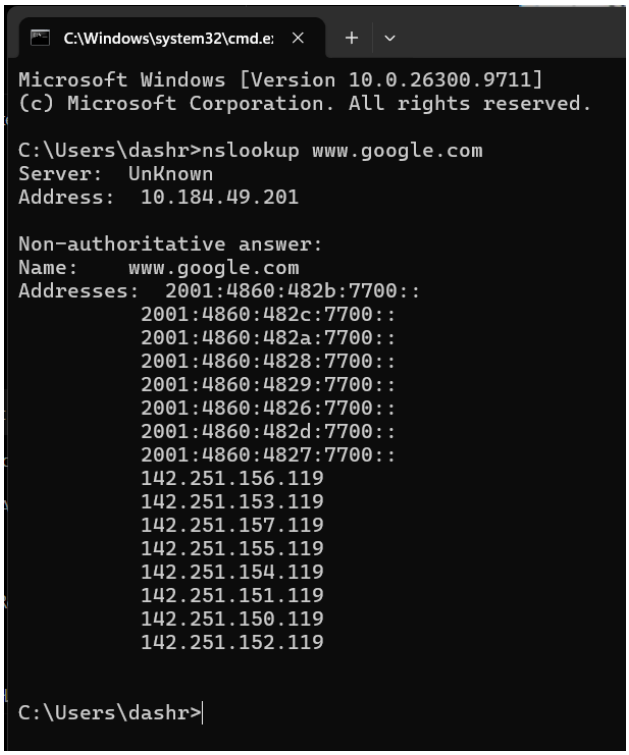
Setting	Value
Preferred DNS Server	8.8.8.8
Alternate DNS Server	8.8.4.4

5. Click OK to save the settings.
6. Open Command Prompt.
7. Execute the following command:
   ipconfig /flushdns



This command clears the locally stored DNS cache.

- **Step 7: Verify the Solution**
- Open Command Prompt.
- Execute the following command:
  nslookup www.google.com
- Check whether the DNS server returns an IP address.
- Now open the web browser and enter www.google.com.
- Verify that the website opens successfully.



- **Troubleshooting Summary Table**

Problem	Possible Cause	Solution
Website does not open by name	DNS server problem	Configure a valid DNS server.
DNS request timed out	DNS server is not responding	Check server connectivity and DNS service.
Incorrect DNS address	Wrong network configuration	Enter the correct DNS server IP.

Old DNS records	Outdated local DNS cache	Run ipconfig /flushdns.
IP works but domain fails	DNS resolution failure	Test and correct DNS configuration.

- **Result**
- Successfully diagnosed a DNS resolution problem using the nslookup command. The network connectivity was checked, the DNS server configuration was verified, and the DNS settings were corrected. After applying the solution, domain names could be resolved into their corresponding IP addresses.

❖ **4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Check for network connectivity, DHCP scope exhaustion, and server availability.**

➢ **Aim**

➢ To identify and troubleshoot a DHCP-related network problem in which a client device is unable to obtain an IP address from the DHCP server and automatically receives an APIPA address.

➢ **Introduction**

- DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network configuration details to devices connected to a network.
- APIPA stands for Automatic Private IP Addressing. When a Windows computer cannot contact a DHCP server to obtain an IP address, it may automatically assign itself an IP address from the range:

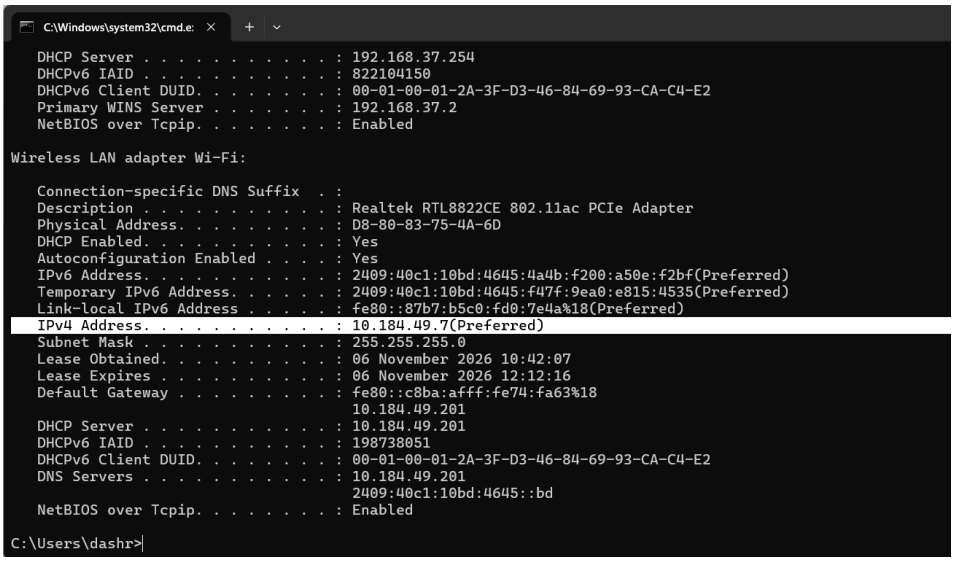
169.254.0.1 to 169.254.255.254

- The subnet mask is generally 255.255.0.0.
- An APIPA address usually indicates that the device could not obtain an IP address from a DHCP server. As a result, the device may communicate with other devices on the same APIPA network but generally cannot access other networks or the internet.

- **Possible Causes of the Problem**
- The following are the common causes of DHCP failure:
- **1. Network Connectivity Problem**
- Ethernet cable may be disconnected or damaged.
- WiFi connection may be unavailable.
- Switch or router may be powered off.
- Network interface may be disabled.
- **2. DHCP Scope Exhaustion**
- The DHCP server may have no available IP addresses left in its address pool.
- All available IP addresses may already be assigned to other devices.
- The DHCP scope may be configured with a very small address range.
- **3. DHCP Server Unavailability**
- The DHCP server may be switched off.
- DHCP service may be stopped or disabled.
- The server may have a network connectivity problem.
- The DHCP server may not be responding to client requests.
- **4. Incorrect DHCP Configuration**
- DHCP scope may have an incorrect IP address range.
- Subnet mask may be configured incorrectly.
- DHCP scope may be disabled.
- Incorrect VLAN configuration may prevent communication with the DHCP server.
- **5. DHCP Relay Agent Problem**
- In networks with different subnets, a DHCP relay agent may be required.
- Incorrect relay configuration can prevent DHCP requests from reaching the server.

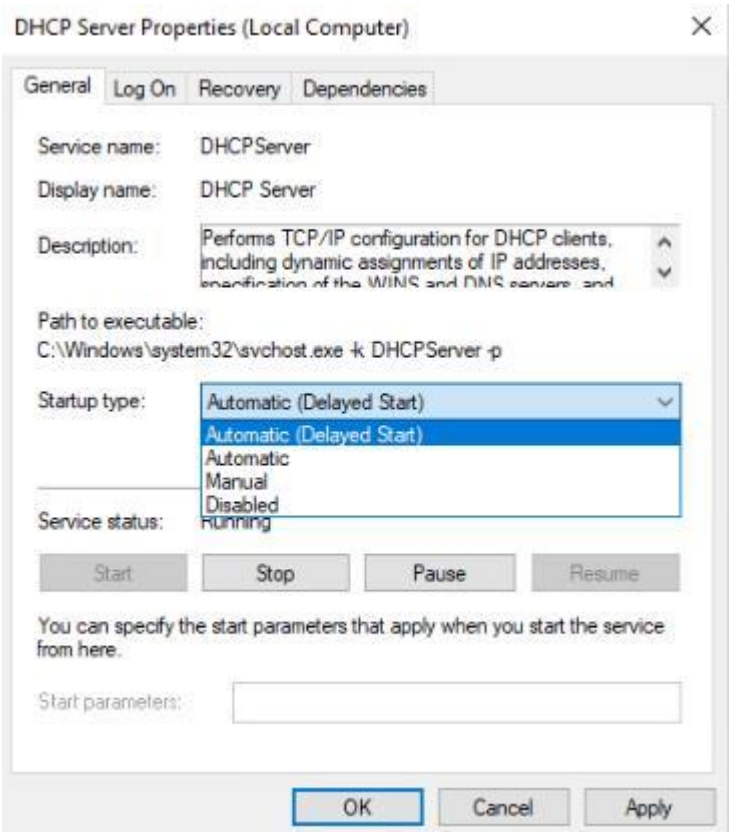
- **Requirements**
  1. Windows computer or laptop.
  2. Cisco Packet Tracer or GNS3.
  3. DHCP server.
  4. Router and network switch.
  5. Client PC.
  6. Ethernet cables.
  7. Command Prompt (CMD).

- **Step-by-Step Troubleshooting Process**
- **Step 1: Check the IP Address of the Client Device**
- Open Command Prompt on the client computer.
- Execute the following command:
  ipconfig /all
- Press Enter.
- Check the IPv4 Address field.
- If the address starts with 169.254, the device has received an APIPA address.



- **Step 2: Check Network Connectivity**
- Check whether the Ethernet cable is properly connected.
- Verify that the network switch and router are powered on.
- Check whether the network adapter is enabled.
- If using WiFi, verify that the device is connected to the correct wireless network.
- Check the network connection status.
- You can also use:
- ping 127.0.0.1
- This command checks the local TCP/IP networking stack.
- **Step 3: Check DHCP Server Availability**
- Open the DHCP server.
- Verify that the server is switched on.
- Check whether the DHCP service is running.
- Verify that the server has a valid IP address.
- Check whether the DHCP server is connected to the network.

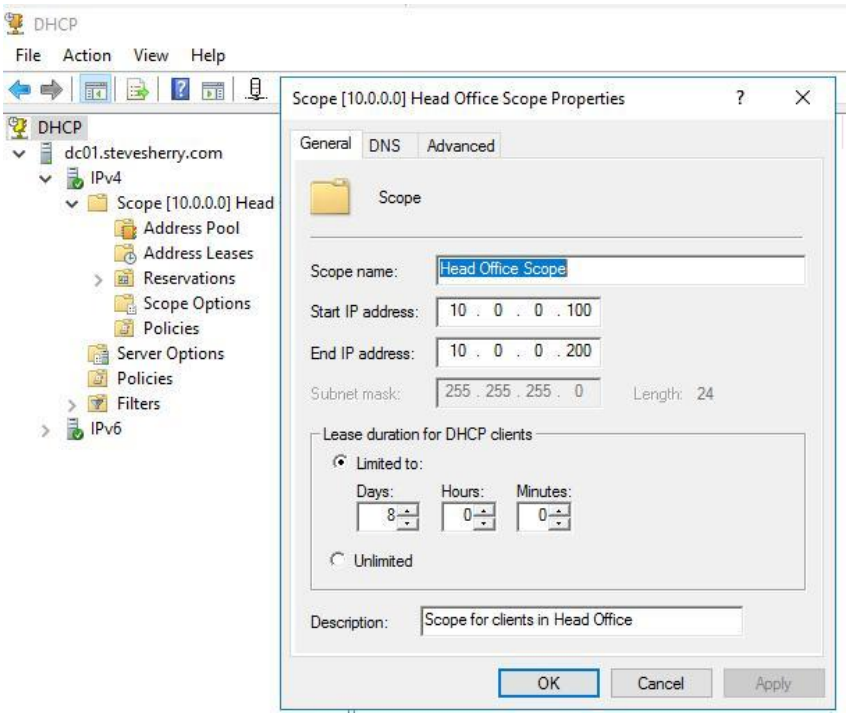
- On a Windows DHCP server, the service can be checked through:
- services.msc
- Find the DHCP Server service and verify its status.



- **Step 4: Check DHCP Scope Exhaustion**
- A DHCP scope defines the range of IP addresses that can be assigned to client devices.
- For example:

DHCP Setting	Configuration
Starting IP	192.168.1.10
Ending IP	192.168.1.20
Subnet Mask	255.255.255.0
Available Addresses	11

- Open the DHCP server configuration.
- Check the available IP address pool.
- Verify whether all IP addresses have already been assigned.
- Remove expired or unnecessary leases if appropriate.
- Increase the DHCP address range if more addresses are required and the subnet has available addresses.



- **Step 5: Check DHCP Configuration**
- Open the DHCP server settings.
- Verify that the DHCP scope is enabled.
- Check the starting and ending IP addresses.
- Verify the subnet mask.
- Check the default gateway and DNS server settings.
- Ensure that the DHCP server is configured for the correct network.
- Example correct configuration:

Setting	Value
Start IP Address	192.168.1.10
End IP Address	192.168.1.100

Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

- **Step 6: Renew the IP Address**
- After correcting the network or DHCP server problem, open Command Prompt on the client computer and execute:
- ipconfig /release
- This command releases the current IP configuration.
- Now execute:
- ipconfig /renew
- This command requests a new IP address from the DHCP server.
- Finally, verify the configuration using:
- ipconfig /all
- **Step 7: Test Network Connectivity**
- After receiving a valid IP address, test the connection to the DHCP server or router.
- Execute:
- ping 192.168.1.1
- If the router responds successfully, local network connectivity is working.
- Then test external connectivity if the network has internet access:
- ping 8.8.8.8
- If the tests are successful, the DHCP-related problem has been resolved.

- **Troubleshooting Summary Table**

Problem	Possible Cause	Solution
APIPA address (169.254.x.x)	DHCP server not responding	Check DHCP server availability.
No network connection	Damaged cable or disabled adapter	Check cable and enable adapter.
No available IP addresses	DHCP scope exhausted	Increase the address pool or free unused leases.
DHCP service stopped	Server service unavailable	Start the DHCP service.
Incorrect IP configuration	Wrong DHCP scope settings	Correct the DHCP configuration.
DHCP request not reaching server	VLAN or relay issue	Check VLAN and DHCP relay configuration.

- **Result**
- Successfully identified the possible causes of DHCP failure when a client device receives an APIPA address. Checked network connectivity, DHCP server availability, and DHCP scope capacity. After correcting the configuration and renewing the IP address, the client received a valid IP address from the DHCP server.

❖ **5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.**

➢ **Introduction**

➢ Network services such as DNS (Domain Name System) and DHCP (Dynamic Host Configuration Protocol) play an important role in computer networking.

- **DNS (Domain Name System):** DNS translates human-readable domain names, such as www.google.com, into IP addresses so that computers can communicate with servers.
- **DHCP (Dynamic Host Configuration Protocol):** DHCP automatically assigns IP addresses and other network configuration details, such as subnet mask, default gateway, and DNS server address, to devices on a network.

➢ Sometimes these services may not work properly due to incorrect configurations, server failures, or network connectivity problems. Troubleshooting helps identify and resolve these issues.

➢ **Common DNS Issues and Their Solutions**

➢ The following table describes three common DNS issues, their symptoms, possible causes, and recommended solutions.

No.	DNS Issues	Symptoms	Possible Causes	Recommended Solutions
1	DNS Server Not Responding	Websites cannot be opened using domain names, and the browser may display a DNS server not responding error.	DNS server is unavailable, network connectivity problems, or incorrect DNS server settings.	Check network connectivity, verify DNS server settings, restart the DNS service if authorized, or use a valid DNS server.
2	Incorrect DNS Configuration	Users are unable to access websites or are redirected to incorrect IP addresses.	Incorrect DNS server IP address, wrong DNS records, or misconfigured DNS settings.	Verify DNS server configuration, correct the DNS records, and ensure that the correct DNS server address is configured.
3	DNS Cache Problems	Recently changed websites may not open correctly, or users may receive an outdated IP address.	Outdated or corrupted DNS cache stored on the computer.	Clear the DNS cache using the ipconfig /flushdns command and try accessing the website again.

- **3. Common DHCP Issues and Their Solutions**
- The following table describes three common DHCP issues, their symptoms, possible causes, and recommended solutions.

No.	DHCP Issues	Symptoms	Possible Causes	Recommended Solutions
1	DHCP Server Not Available	A computer cannot obtain an IP address automatically and may receive an APIPA address starting with 169.254.	DHCP server is offline, DHCP service is stopped, or network connectivity is unavailable.	Check the DHCP server status, restart the DHCP service if authorized, and verify network connectivity.
2	IP Address Conflict	Two devices may experience network connectivity problems because they are using the same IP address.	Duplicate static IP addresses, incorrect DHCP scope configuration, or overlapping IP address ranges.	Identify the conflicting devices, remove duplicate IP configurations, and configure the DHCP scope correctly.
3	DHCP Scope Exhaustion	New devices cannot connect to the network or obtain an IP address automatically.	All available IP addresses in the DHCP scope have been assigned to devices.	Increase the DHCP address pool, remove unused leases, or adjust the DHCP scope to provide more available IP addresses.

- **Important Troubleshooting Commands**
- The following commands are useful for identifying and resolving common DNS and DHCP problems in Windows.

Command	Purpose
ipconfig /all	Displays complete IP configuration, including DNS and DHCP information.
ipconfig /flushdns	Clears the DNS resolver cache.
ipconfig /release	Releases the currently assigned DHCP IP address.
ipconfig /renew	Requests a new IP address from the DHCP server.
nslookup google.com	Checks DNS name resolution for a domain.
ping 8.8.8.8	Tests basic network connectivity to an IP address.